



UMS
UNIVERSITI MALAYSIA SABAH

KP24302 INTERACTION DESIGN [1-2023/2024]

LECTURER : DR. EDMUND NG GIAP WENG

INDIVIDUAL ASSIGNMENT 1

SMART FUNERAL APPLICATION PROPOSAL

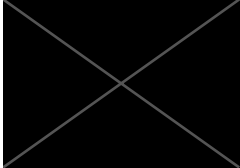
NO	NAME	MATRIC NUMBER
1.	LEENA A/P SELLAMANY	
2.	SHWEHTTA A/P PARANI KUMAR	

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1.0 FEATURES

1.1 USER AUTHENTICATION

An essential feature of the funeral app's security framework is user authentication, which serves to safeguard sensitive information and ensuring that only authorized individuals have access to certain features such as managing profiles, planning events, and engaging in digital memorialization. This is executed by ensuring users create secure login credentials, usually a combination of complicated password and distinctive usernames. Secure passwords can be encouraged by requiring passwords with a mix of lower and uppercase letters and must at least include 1 symbol or number. Two Factor Authentication further adds an extra layer of security. This entails emailing the user a verification code to their registered mobile device or email address, making sure that access still requires a step beyond compromising only of login credentials. Biometric authentication adds an extra layer of security by relying on unique physiological or behavioral traits, such as fingerprints or facial features. This significantly reduces the risk of unauthorized access, ensuring that sensitive information related to memorial events, profiles, and digital memorials remains secure. Fingerprint or facial recognition provides a seamless and user-friendly login experience. Users are spared the need to remember and input complex passwords during times that may be emotionally challenging.

1.2 PROFILE MANAGEMENT

The Profile Management feature within the funeral app provides users with a structured mechanism to create and administer profiles, encompassing not only their own accounts but also extending this capability to include profiles for deceased individuals. They can upload photographs and other visual content, share narrative elements, and methodically enter and edit relevant details to create customized profiles. The platform also has privacy settings, giving users the necessary control over the visibility and accessibility parameters of the profiles. The Profile Management feature is a practical way for users to keep accurate digital records. In essence, Profile Management is a professional utility that facilitates the organized and secure management of information related to individuals, acknowledging both the living and deceased within a structured and controlled framework. The combination of AR and IoT makes the Profile Management feature a dynamic, ever-evolving environment. New memories can be added to user profiles, and these updates are automatically reflected in connecting IoT-enabled objects and 3D holographic displays. By enabling users to virtually position digital artifacts within the holographic environment, augmented reality features can improve profile personalization by enabling users to create visually expressive and customizable profiles. By utilizing Internet of Things connectivity, the application can recommend particular memories or significant events based on user interactions and preferences, making profile management more adaptive and user-centric profile management experience.

1.3 LEGACY VIDEO MESSAGES

The Legacy Video Messages feature is a distinctive element of the funeral app, providing users with ways to create and preserve personal sentiments for future occasions. This feature empowers users to record video messages and also write letters. This feature also allows the users to schedule the release of these messages on specific dates or events. These video messages can serve as a lasting legacy, offering a more immersive and emotionally resonant connection between the individual leaving the message and their intended recipients. Practically, this feature ensures that users can plan for the future dissemination of their recorded messages, aligning with significant milestones, anniversaries, or events. In order to comprehend the emotional context and sentiments expressed, the AI update can examine user-shared text and media content. Written notes, Legacy Video Messages, and any other shared content pertaining to condolences or remembrance are included in this analysis. By using NLP, the AI is able to produce emotionally meaningful and context-aware responses. It can interpret linguistic nuances, so suggested messages of sympathy are not only grammatically correct but also sound sincere and caring. As a result of user interactions and preferences, the AI gradually gains the ability to deliver more relevant and personalized condolence messages. Through adaptive learning, the AI's responses are guaranteed to be in line with the various ways that people express and manage their grief.

1.4 DIGITAL MEMORIALIZATION

The Digital Memorialization feature is a cornerstone of the funeral app, offering users a dedicated space to commemorate their loved ones in a digital realm. This feature makes it easier to create a digital memorial where people can share heartfelt films, priceless photos, and poignant stories. One important component of this feature could be the possible addition of a virtual memorial wall. This acts as a communal canvas on which loved ones can post condolence and encouragement notes. The virtual memorial wall serves as a focal point for messages of compassion and unity, fostering conversations and serving as evidence of the deceased person's legacy. With augmented reality, it is possible to create customized avatars or pictures of the deceased for use in the virtual environment. By reacting dynamically to user interactions, these avatars can give users a feeling of companionship and virtual presence. Make using the AR Memorials more of a team effort. With AR devices, friends and family can join the virtual space at the same time, no matter where they are in the world. Users can leave AR-enhanced messages, including multimedia elements like 3D virtual objects, photos, or videos, on an AR-enhanced virtual memorial wall. By assigning these messages to particular points in the virtual environment, an interactive and spatialized memorial wall can be made.

1.5 EVENT PLANNING

The Event Planning feature of the funeral app is designed to simplify and enhance the organization of memorial events, ensuring a seamless coordination of attendees and friends. Users can utilize forms, polls, and calendars to efficiently determine the most suitable date and time that accommodates the majority, thus optimizing participation. Forms are incorporated to allow users to gather preferences and availability, guaranteeing that the selected event schedule coincides with participant convenience. Polls are a flexible means of obtaining rapid input, which facilitates group decision-making when choosing dates or event-related options. Calendars are essential for finalizing and visualizing the schedule of events. The dates with the greatest level of agreement among prospective attendees are readily referenced and chosen by users. Reminders and notifications are included in the feature to improve user participation and attendance. It is ensured that attendees are informed about the event details and any potential changes by providing them with timely updates. By proactively recommending dates and times that coincide with the availability of the majority of participants, the AI-driven scheduling system seeks to optimize event attendance. Beyond basic availability checks, this analysis looks into things like favorite times of day, ongoing obligations, and past attendance trends. This guarantees that the timetable selected accommodates the needs of the greatest number of participants, resulting in an event that is both meaningful and well-attended.

1.6 YEARLY ANNIVERSARIES

The Yearly Anniversaries feature in the funeral app allows users to proactively commemorate the passing of a loved one by setting up personalized reminders and notifications for significant yearly milestones. On the anniversary date, the app provides a range of thoughtful features to facilitate a meaningful remembrance. The inclusion of virtual candle lighting or lighting incense is a poignant aspect of this feature. Users can virtually light candles or incense stick in memory of the departed, fostering a symbolic and respectful gesture. Sending virtual flowers is another feature that enhances the online commemoration experience. To further enhance the user experience, the app incorporates the option for virtual meetings online. This feature enables family and friends to come together in a virtual space on the anniversary date, fostering a sense of unity and support. Attending memorial events in virtual reality (VR) offers users an immersive and lifelike experience, just like being there in person. Attendees of VR events can view presentations, interact with the surroundings, and engage in more emotionally impactful interactions. Add VR event spaces to the Event Planning feature so that users can organize and conduct virtual memorial events. These areas can be tailored to the mood and importance of the event, bringing together participants who are physically apart.

2.0 FUNCTIONS

2.1 USER AUTHENTICATION

- **create_user_account(username, password, email)**
- **create_user_account(username, password, phone number)**
 - Create a new user account with a unique username, strong password, and email.
- **authenticate_user(username, password)**
 - Verify username and password against stored credentials.
- **generate_verification_code()**
 - Generate a unique verification code for two-factor authentication.
- **reset_password(username, new_password)**
 - Reset the user's password.
- **enable_two_factor_auth(user_id)**
 - Enable two-factor authentication for a user.
- **disable_account(user_id)**
 - Disable a user account for security reasons.

2.2 PROFILE MANAGEMENT

- **create_profile(user_id, profile_data)**
 - Create a new user profile with provided data.
- **update_profile(user_id, updated_data)**
 - Update user profile with new information.
- **retrieve_profile(user_id)**
- **search_profile(user_id)**
 - Retrieve user profile information.
- **delete_profile(user_id)**
 - Delete the user's profile and associated data.
- **change_privacy(user_id, privacy_options)**
 - Modify user privacy settings.

- **upload_profile_pic(user_id, pic_data)**
 - Allow users to upload or change their profile picture.
- **share_profile(user_id, recipient_id)**
 - Share a user's profile with another user.

2.3 LEGACY VIDEO MESSAGES

- **record_legacy_message(user_id, message_type, content, schedule_date)**
 - Record a video message or write a letter, with an option to schedule its release.
- **schedule_legacy_message(user_id, message_id, schedule_date)**
 - Schedule the release of a legacy message on a specific date.
- **edit_legacy_message(user_id, message_id, new_content)**
 - Edit a previously recorded legacy message.
- **delete_legacy_message(user_id, message_id)**
 - Delete a legacy message.
- **share_legacy_message(user_id, recipient_id, message_id)**
 - Share a legacy message with another user.
- **analyze_sentiments_ai(message_content)**
 - Analyze sentiments in the provided message content.

2.4 DIGITAL MEMORIALIZATION

- **create_digital_memorial(user_id, memorial_data)**
 - Create a digital memorial for the user with provided data.
- **add_media(user_id, media_type, media_data)**
 - Add various types of media to the digital memorial.
- **customize_avatar(user_id, avatar_data)**
 - Customize avatar appearance and accessories.
- **recommend_memorial_content(user_id)**
 - Use AI to recommend additional content for the memorial.
- **notify_on_memorial_anniversary(user_id, memorial_id)**
 - Send notifications for upcoming memorial anniversaries.

2.5 EVENT PLANNING

- **plan_memorial_event(user_id, event_details)**
 - Assist in planning memorial events using forms, polls, and calendars.
- **create_event_invitations(user_id, event_details, invitees)**
 - Send invitations to specific users for a memorial event.
- **cancel_event(user_id, event_id)**
 - Allow users to cancel a planned memorial event.
- **view_event_participants(event_id)**
 - Retrieve a list of participants for a memorial event.
- **suggest_optimal_event_date(user_id, participant_availability)**
 - Analyze participant availability and suggest an optimal event date.
- **send_event_reminders(user_id, event_details)**
 - Send reminders and notifications to event participants.
- **share_event_photos(user_id, event_id, photo_data)**
 - Allow users to share photos from memorial events.

2.6 YEARLY ANNIVERSARIES

- **commemorate_anniversary(user_id, anniversary_type, virtual_options)**
 - Set up personalized reminders for yearly milestones.
- **virtual_meeting_in_vr(user_id, meeting_date)**
 - Organize a virtual meeting in VR for a specific date.
- **send_virtual_flowers(user_id, recipient_id, flower_type)**
 - Send virtual flowers as a gesture of remembrance.
- **customize_virtual_meeting_environment(user_id, meeting_settings)**
 - Allow users to customize the virtual meeting environment.
- **attend_virtual_event(user_id, event_id)**
 - Mark a virtual event as attended.

3.0 Notes on the conceptual design and evidence gathering

Based on the survey we have done in the form of google form regarding the Smart Funeral Application, we have gathered around 30 responses answering the google form. A majority of 83.3% of respondents prefer mobile application to access the Smart Funeral Application compared to a web-based platform. Hence, it is assumed that mobile application is easier to access and engage than using a web-based platform because web-based platform requires Internet access making it complicated to access the application anytime anywhere to keep updated about the upcoming event. Meanwhile, mobile app serves the best in a sense that it mostly does not require Internet access. Therefore, it will be even more convenient for elderly people or those who cannot get easy access of Internet to use the application at ease.

Among the features that we would like to include on our Smart Funeral Application are User Registration and profile management, legacy messages, digital memorialisation, event planning and yearly anniversaries. Event planning is considered as one of the features that the respondents have found the most helpful compared to others. This is because it is assumed that event planning has become very hard these days as people are busy with their schedules, hence it is difficult and sometimes burdensome to plan an event physically such as booking place, food and etc to cater the guests for the event for the deceased ones. Moreover, user registration and profile management feature is also found helpful because it should be the main feature when using an application as it will enable the users to manage their own profile.

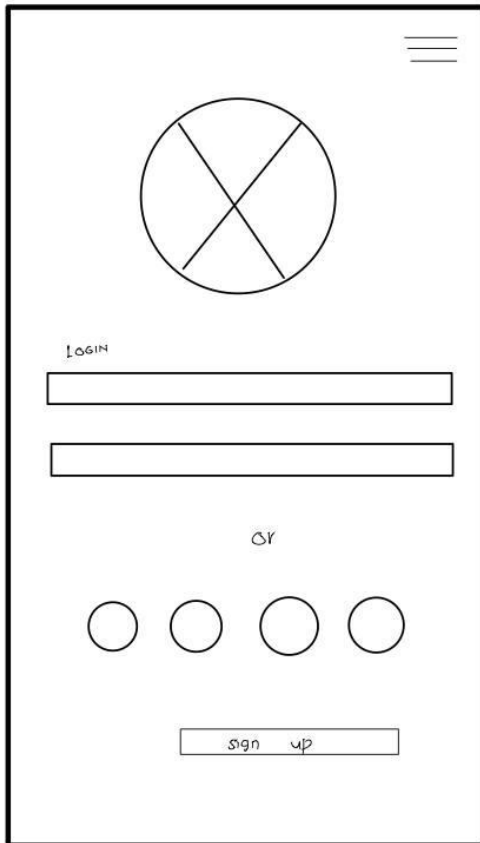
Furthermore, among the futuristic updates that people might expect from a Smart Funeral Application are Virtual Reality Memorials, Augmented Reality Memorials, AI assisted grief counseling, collaborative memorial projects and environmental project considerations. We assume that virtual reality memorial is much expected from such an app as it has a majority of 70% of responses because it makes people to attend event with VR technology. Besides, environmental impact considerations and collaborative memorial projects are also considered the much expected futuristic updates needed for the app. As of the environmental consideration feature, it allows the users to choose among the eco-funeral funeral options such as sustainable burials. Therefore, it grows awareness among people about saving the environment for the future generation. Apart that, collaborative memorial projects is another feature that is assumed to be a

futuristic update that is expected to include on the app. This is because the feature enable users to share photos and videos of the deceased ones as it could be a tribute for them as a sign of remembrance and respect for the deceased ones.

The information that users would like to be able to access or share through the app is media such as photos or video. We assume that it is because media is the one that keeps the deceased ones alive as we can all look through the media and reminisce the memories left behind by the deceased ones. It is followed by condolences messages because it is the most respectable way to mourn for the deceased one as the users of the app can share their messages that would be heart melting for the family and friends of the deceased ones. As per the safety measures, privacy setting has the most percentage of responses as a mean to ensure the privacy of sensitive information. Since privacy setting enables the users to choose whether to keep their profile or public, it ensures that the information regarding the user is not misused.

As per the method to receive funeral details, users prefer different types of methods to receive such details including notifications, phone calls, reminder, emails and Whatsapp messages. Among all the methods mentioned, notification is the most preferred method as it is more convenient for the users. Lastly, in order to handle post-funeral services, users prefer notification for anniversaries. This is because the notification for anniversaries is useful to remind others about the death anniversaries.

4.0 PROTOTYPE SKETCHES



LOGIN

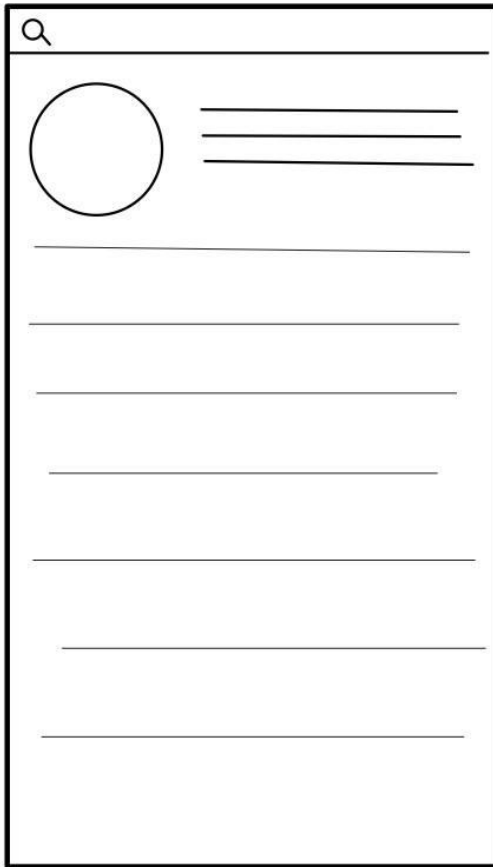
- Enables users to login using their username and password
- Enables new users to sign up to the app

REGISTER

OR

REGISTER

- Enables users to register their accounts using their email address, phone number or google account.



USER PROFILE

- User profile consists of their profile picture and about them
- Includes a search option to search for other users as well

The wireframe shows a vertical layout with several functional areas:

- CALENDAR / POLLS / POSTERS:** A large rectangular area at the top left.
- YEAR:** Three stacked boxes, each containing a grid of 12 squares (3 rows by 4 columns), likely representing a calendar year.
- CHATBOX:** Two speech bubble icons pointing to rectangular input fields.
- POLLS / FORMS:** A rectangular area below the chatbox.
- POSTER:** A larger rectangular area below the polls/forms section.
- PARTICIPANTS ID:** A list of five radio buttons, each followed by a horizontal line for a name or ID.
- VENUE / NOTIFICATIONS / ANNOUNCEMENT:** A rectangular area at the bottom right.

EVENT PLANNING

- Consists of calendar to mark the event date
- Chatbox to discuss about the event they are planning
- Includes poster of the event
- Participants ID is also included in the list of guests showing up for the event
- Poll/form column is to used to ask opinion on the date,venue and other details when planning the event

Flowers

Commemoration

meeting

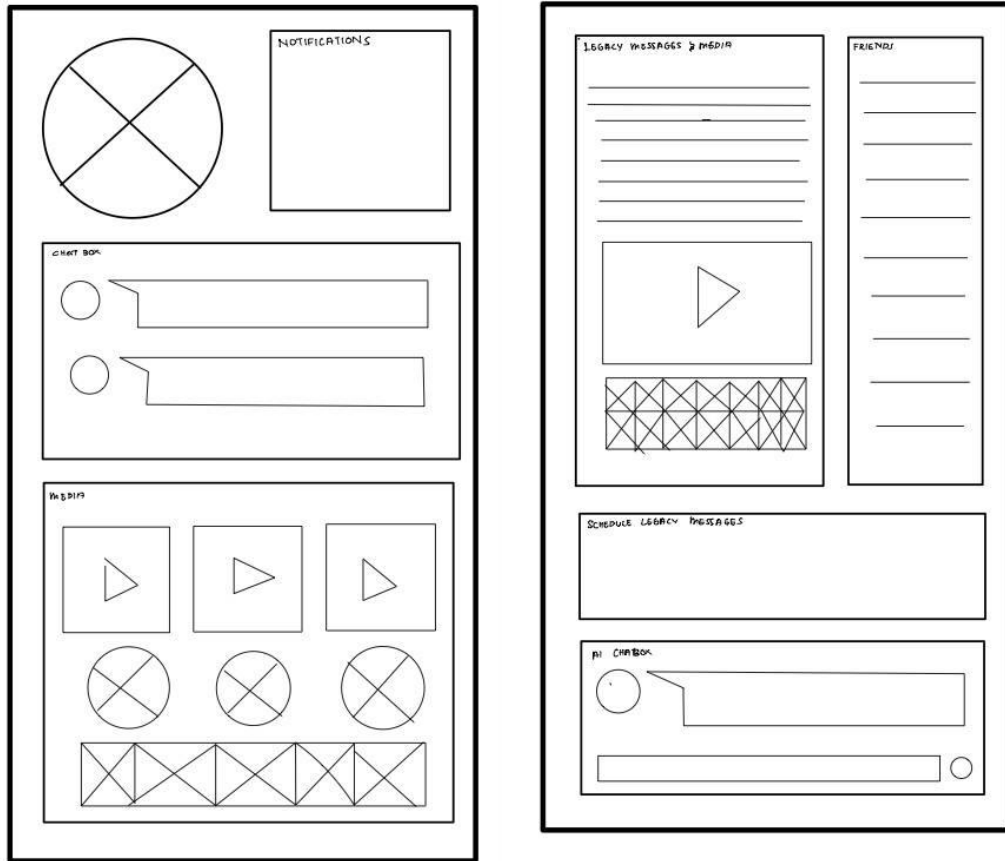
Join

VR session

Join

YEARLY ANNIVERSARY

- Includes flowers column where the guest for the anniversary will give virtual flowers for the deceased ones
- Commemoration box is to allow the guests to celebrate the deceased ones by writing messages about them
- Meeting link is provided to join the yearly anniversary virtually
- Includes VR session of the deceased ones



- Notification reminder is given as well as chat box followed by video messages and photos
- For legacy messages and videos, the media and friends list of the deceased one are included
- Schedule legacy message option is also included as well as AI chatbox to communicate with an AI tool regarding the details of the event

5.0 PROTOTYPE EVALUATIONS AT EACH DESIGN ITERATION

5.1 USER AUTHENTICATION

Iteration 1:

- Test the basic authentication process to ensure users can securely log with username/password authentication
- Conduct a basic security audit

Iteration 2:

- Introduce and test the effectiveness of multi-factor authentication methods such as message codes and one-time password
- Implement and test any password policies such as the password requirements (eg : combination of uppercase & lowercase)
- Test the account recovery process

Iteration 3:

- Introduce and assess the effectiveness of biometric authentication methods
- Gather user feedback on the effectiveness of biometric authentication
- Implement any additional security enhancements
- Conduct a thorough security review
- Ensure that all authentication methods are accessible to users

Final Iteration:

- Conduct a comprehensive security audit to ensure the overall security of the authentication
- Ask users about the perceived security, ease of use, and effectiveness of the authentication methods.
- Ensure that user documentations related to authentication are clear

5.2 PROFILE MANAGEMENT

Iteration 1:

- Evaluate how easily users can create a profile
- Check if the basic features inherent
- Conduct interviews or surveys to gather user feedback on the overall experience

Iteration 2:

- Evaluate the flow for creating and managing profiles for deceased loved ones
- Check if users can transition between their own profile and those of deceased ones
- test the collaborative features
- Ensure that sharing memories and interacting with shared content is straightforward.
- Test the system's response to different types of data input.

Iteration 3:

- assess the emotional impact on users.
- Use qualitative methods like interviews to understand how users feel about the experience.
- Evaluate the security measures in place for protecting sensitive information.
- Test the prototype on different devices

Final Iteration:

- Identify any inconsistencies while using the prototype
- test the prototype's performance with a larger dataset to ensure it can handle a significant number of profiles.
- Collect user feedback on the complete prototype

5.3 LEGACY VIDEO MESSAGES

Iteration 1:

- Test the video recording functionality to ensure users can easily record and save video messages.
- Assess the clarity and quality of recorded videos.
- Evaluate the scheduling interface to set specific dates or events for the release of video messages.
- Gather user feedback on the initial experience

Iteration 2:

- Introduce and assess any editing features for video messages and a preview option before scheduling.
- Introduce and test the notification system for users to be reminded of upcoming scheduled releases.

Iteration 3:

- Gather user feedback on the messages that have been released as per the schedule
- Ensure that the recording, scheduling, and playback features are accessible to users
- Ensure that privacy settings are effective and that users have control over who can access their messages.

Final Iteration:

- Evaluate the end-to-end user journey for recording, scheduling, and experiencing the feature
- Test the system's performance with a larger number of scheduled messages
- Ensure that user documentations are clear and helpful

5.4 DIGITAL MEMORIALISATION

Iteration 1:

- Gather user feedback on the initial experience of creating a digital memorial.
- Test the navigation flow within the digital memorial space

Iteration 2:

- Evaluate improvements made to multimedia content
- Assess the clarity and user-friendliness of the multimedia upload process.
- Test the virtual memorial wall where users can leave messages of condolence and support.
- Introduce and assess features that enrich community interaction

Iteration 3:

- Gather feedback on the community interaction features
- Ensure that all features, including multimedia content and community interaction, are accessible
- Test privacy settings and moderation features

Final Iteration:

- Evaluate the end-to-end user journey
- Test the system's performance with a larger amount of memorial content and interactions
- Ensure that user documentation are clear and helpful

5.5 EVENT PLANNING

Iteration 1:

- Evaluate the ease of use for setting up an event
- Test calendar feature for selecting dates and times.
- Test the navigation flow within the event planning section

Iteration 2:

- Evaluate the effectiveness of using forms and polls for creating an event
- Assess improvements made to the calendar feature
- Ensure that users can select dates and times that work for the majority
- Assess the notification system after it being introduced
- Test the effectiveness of reminders

Iteration 3:

- Assess if users find the reminders helpful and relevant.
- Ensure that all features, including notifications and calendars, are accessible to users

Final Iteration:

- Evaluate the end-to-end user journey for planning an event, from initial setup to receiving notifications.
- Identify any remaining usability issues
- Test the system's performance with a larger number of attendees
- Ask users about the impact of the feature
- Ensure that user documentation are clear and helpful

5.6 YEARLY ANNIVERSARIES

Iteration 1:

- Assess the ease of setting up reminders and notifications for yearly anniversaries.
- Gather user feedback on the initial experience of setting it up.

Iteration 2:

- Evaluate improvements made to the customization options for reminders
- Introduce and assess features like virtual candle lighting, sending virtual flowers, and sharing memories on the anniversary date.
- Test the effectiveness of these features

Iteration 3:

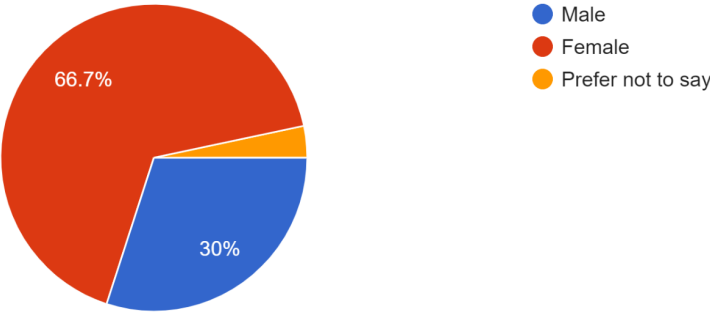
- Gather feedback on the virtual commemorative features
- Ensure that all features, including reminder setup and virtual commemorations, are accessible to users
- Test privacy settings for virtual commemorations

Final Iteration:

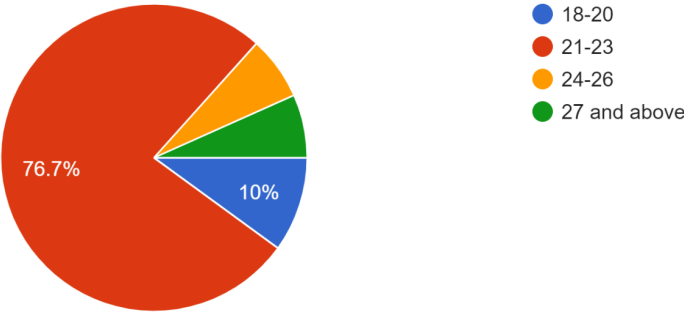
- Evaluate the end-to-end user journey for setting up reminders, receiving notifications, and engaging in virtual commemorations on yearly anniversaries.
- Test the system's performance with a larger number of users

6.0 APPENDIX

Gender
30 responses

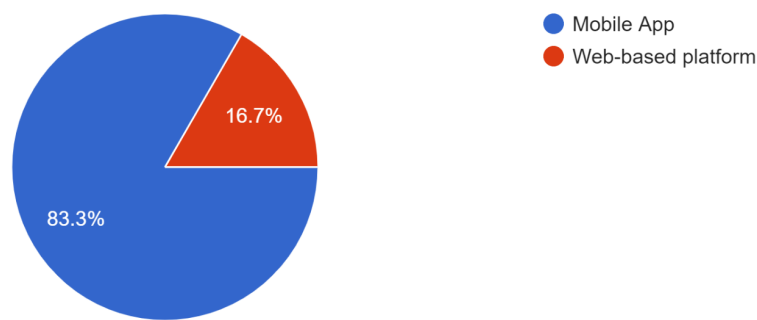


Age
30 responses



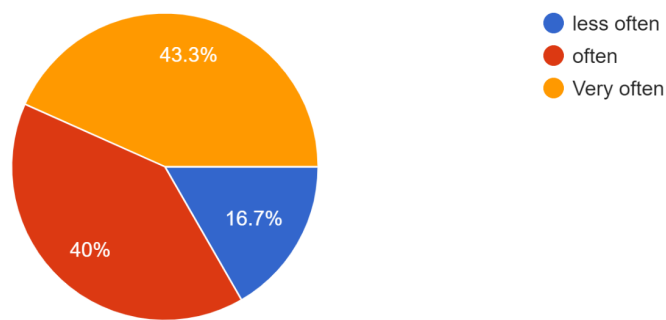
How would you want to access a Smart Funeral Application?

30 responses



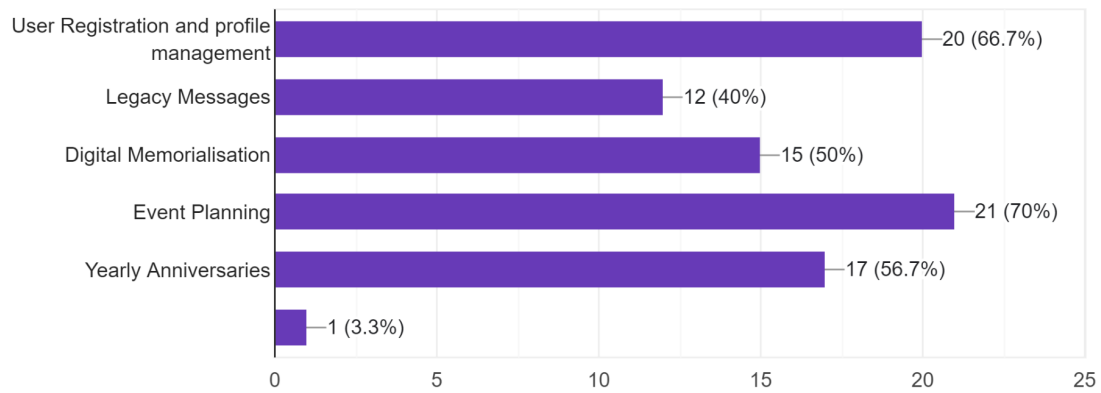
How often do you use mobile app?

30 responses



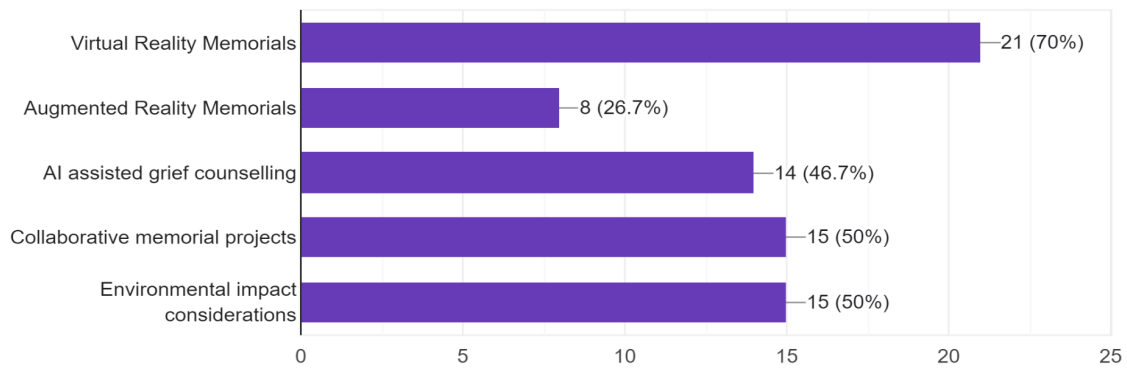
What features would you find the most helpful ?

30 responses



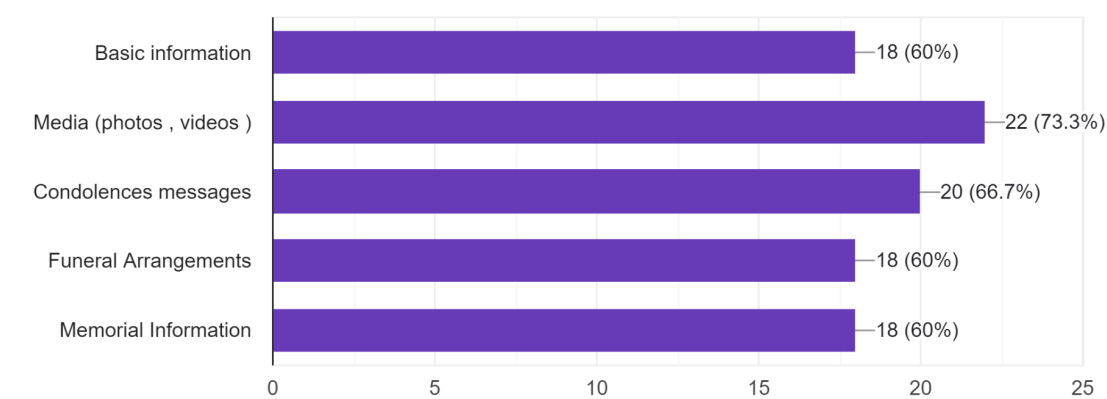
What futuristic updates would you expect from a Smart Funeral Application?

30 responses



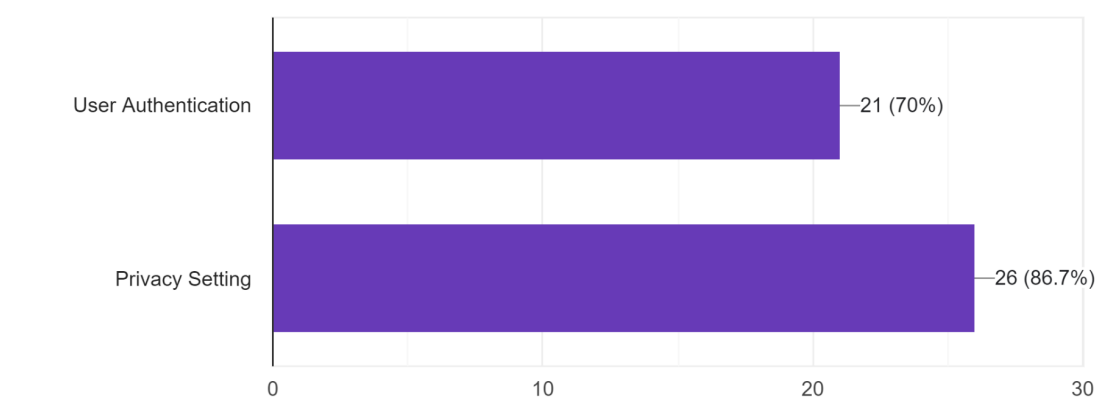
What information would you like to be able to access or share through the app?

30 responses



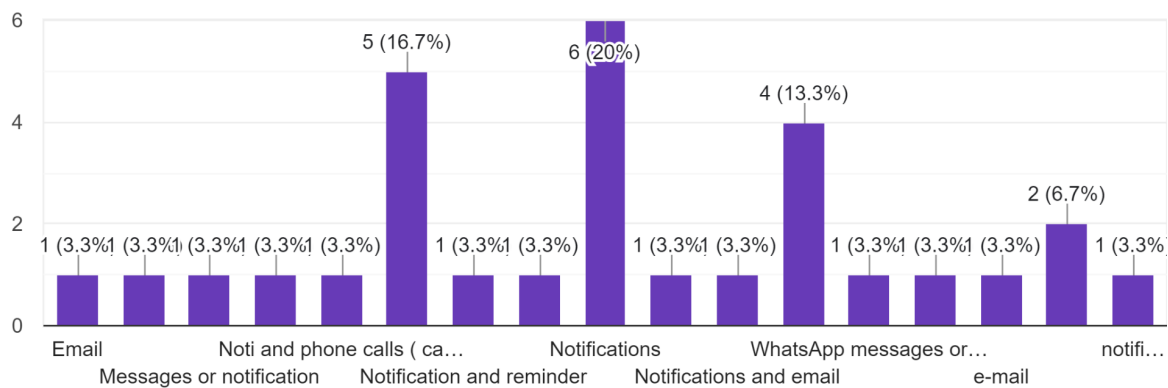
What the safety measures would you expect to ensure the privacy of sensitive information?

30 responses



Which method would you prefer to receive funeral details (notifications ,phone call and etc)?

30 responses



How would you like the app to handle post-funeral services such as memorial or grief support?

30 responses

